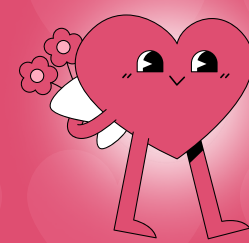
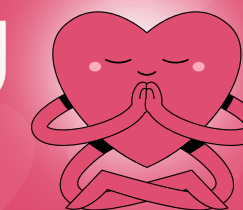


Group 4



Heart Disease Prediction using Machine Learning

KD34403 Machine Learning · Group Assignment · Random Forest



Mohd Asran Bin Lataga | BI23110327
Emely Thu Jia Wei | BI23110306
Shinty Tam Kok Kiong | BI23160407
Karen Lo Xin Xuan | BI23110308
Fidaa' Bin Ahmad Fahmi | BI23110057

Data Insights

Dataset Source

Heart Disease UCI Dataset from UCI Machine Learning Repository Patient records from 4 hospitals: Cleveland, Hungary, Switzerland, Long Beach VA

Dataset Overview

920

Total Samples

16

Raw Features

54.4%

Disease Cases

45.6%

No Disease

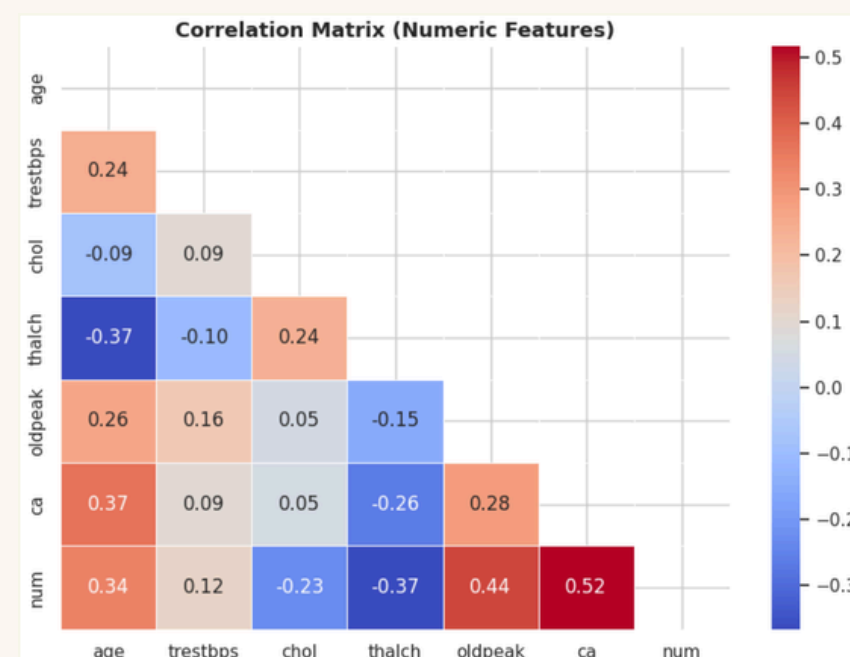
Features

- Numerical: age, trestbps, chol, thalch, oldpeak, ca
- Categorical: sex, cp, restecg, slope, thal, dataset
- Boolean: fbs, exang
- Target: num → 0 (No Disease) / 1 (Disease)

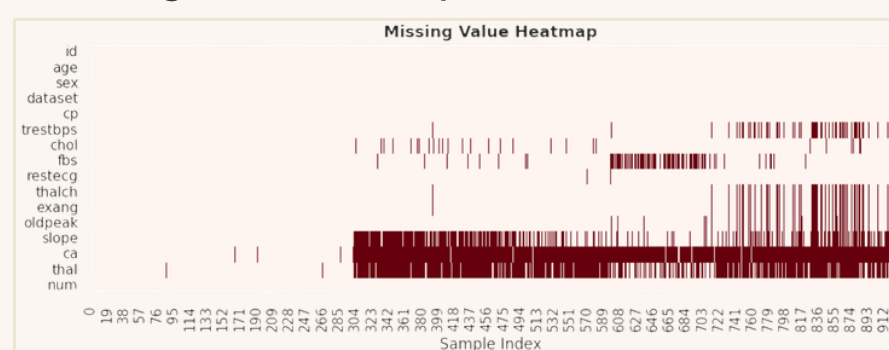
Missing Values (Notable)

- ca: 65.8% missing
- thal: 52.3% missing
- slope: 33.0% missing
- fbs: 9.8% missing

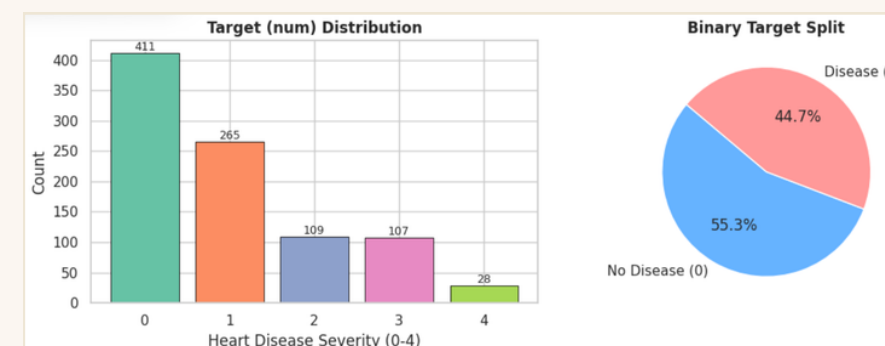
Correlation Matrix



Missing Value Heatmap

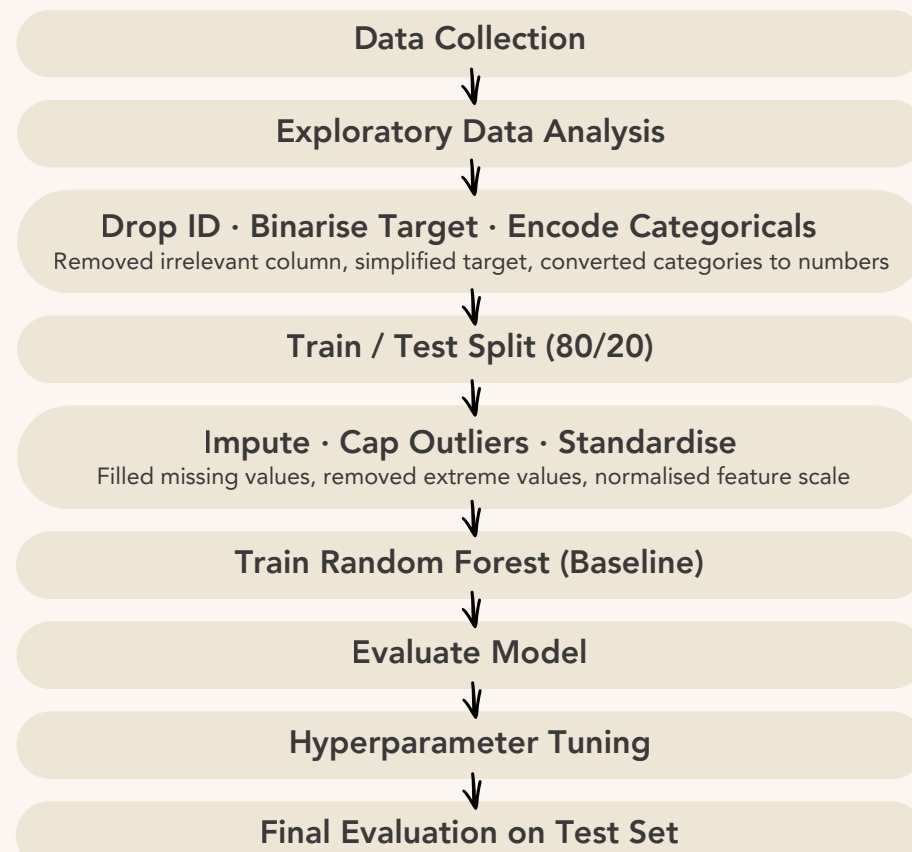


Target Distribution

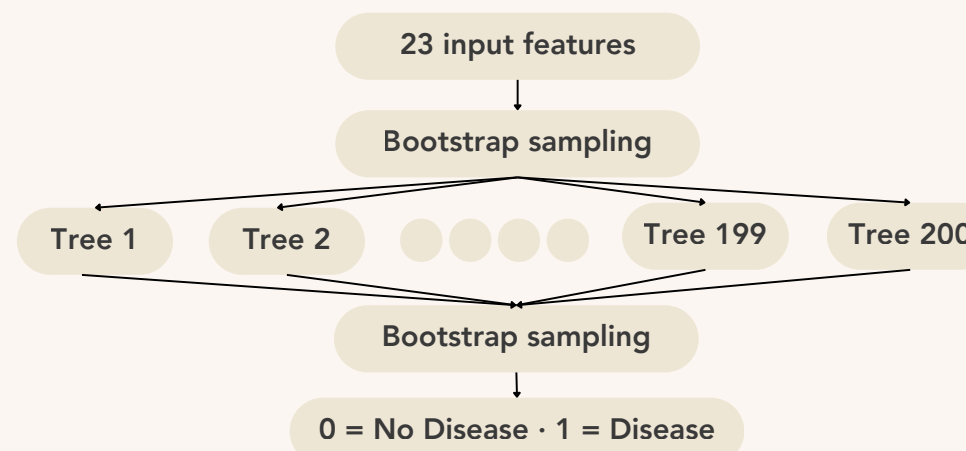


Pipeline & Foundational Logic

Data Flow



Architecture of Random Forest



Why Random Forest Works?

- Tabular data:** handles numeric & categorical features natively
- Small dataset:** bagging creates diverse subsets, no large data needed
- Outlier robust:** threshold-based splits, extreme values don't skew results
- Ensemble:** 200 trees reduce variance over any single decision tree
- Interpretable:** built-in feature importance scores

Training & Evidence

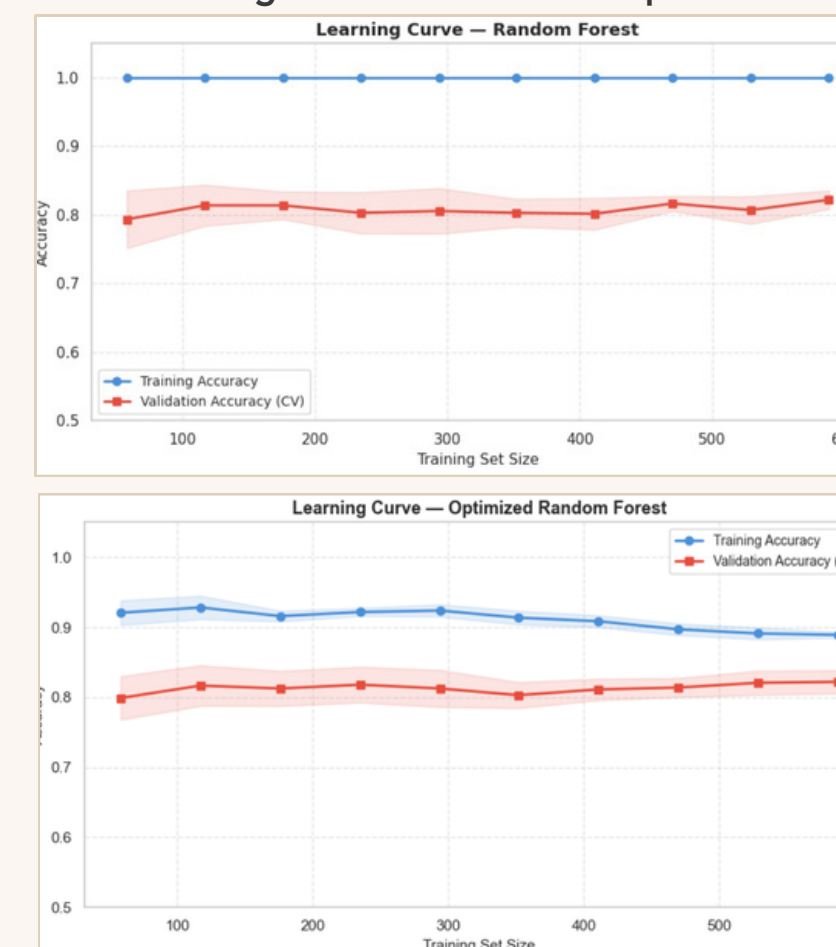
Baseline Model

n_estimators	100
max_depth	None
Train Accuracy	1.00
Test Accuracy	0.853
Test F1	0.872
ROC-AUC	0.927

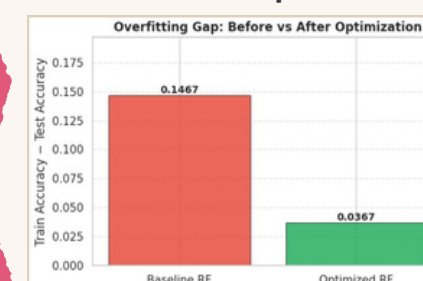
Optimised Model

n_estimators	200
max_depth	8
min_samples_split	10
min_samples_leaf	4
class_weight	balanced

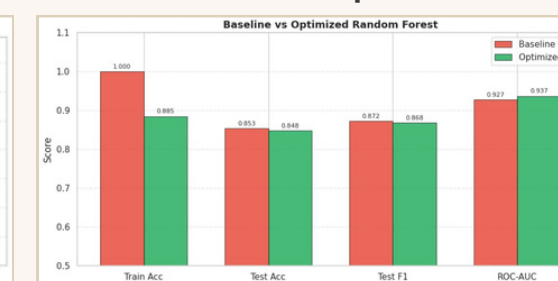
Learning Curve: Baseline vs Optimized



Overfitting Gap: Baseline vs Optimized



Performance Metrics: Baseline vs Optimized



Evaluation & Conclusion

Final Test Results: Optimized Model

84.8%

Accuracy

86.8%

F1-Score

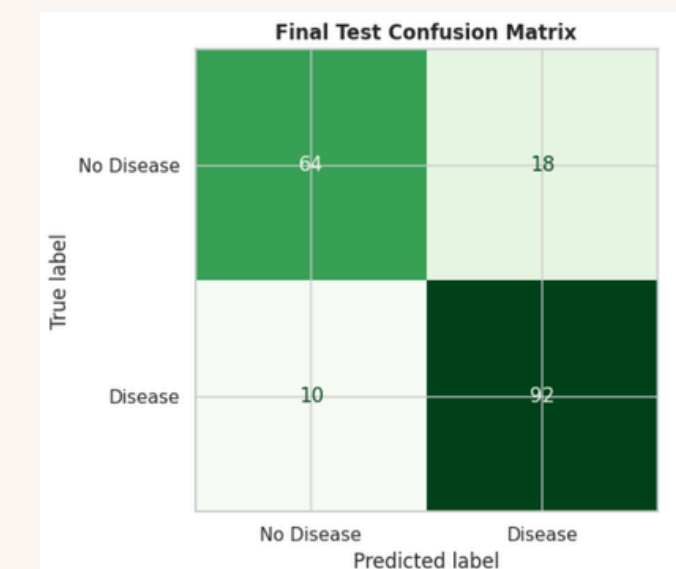
0.937

ROC-AUC

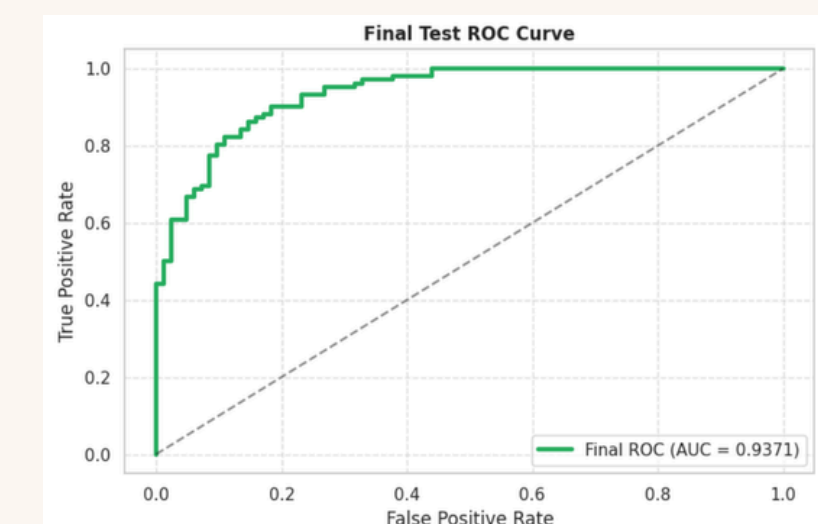
90.2%

Recall

Confusion Matrix



ROC Curve



Error Analysis

False Positives (18):

Healthy patients misclassified as diseased, leading to unnecessary tests.

False Negatives (10):

Actual disease cases were missed, potentially delaying treatment.

Class Imbalance:

The model prioritized recall (90.2%) over precision (84.4%), causing more false alarms.

Conclusion

The optimized Random Forest achieved 84.8% accuracy and 0.937 AUC, with improved generalization after tuning reduced overfitting from 0.147 to 0.037.

Future Work

- Apply SMOTE or cost-sensitive learning to reduce false negatives
- Use feature engineering to improve predictions
- Tune thresholds to increase recall
- Expand the dataset for better generalization

Supporting Materials

Milestone 1 [Click Here](#)

Milestone 4 [Click Here](#)

Milestone 2 [Click Here](#)

Milestone 5 [Click Here](#)

Milestone 3 [Click Here](#)

GitHub [Click Here](#)

